

PAX mounts: before and after

COMPARISON

Before: usual regular post holder. After: rubber-cap-supported assembly below. Labels describe the mounts, not measured damping coefficients.

AFTER: MOUNTING STACK

top to bottom: PAX -> 10mm low profile mount stack constrained by bracing arm -> magnetic base unconstrained floating on -> metal plate floating on -> 3 rubber screw caps placed near the edges of the metal plate, all sitting directly on the -> optical breadboard

MOUNT DESCRIPTION PROVENANCE

Latest comment: same run, but again. checking if strange behavior persists. Stack description comes from 203321, confirmed by the operator; its measurement data are not used.

SOURCE EXPERIMENTS

Before: 2026-09-24/203850_pax-vibration_pax-vibration

After: 2026-09-25/204124_pax-vibration_rubber-tip-base

Both completed; this is offline analysis of the saved raw measurements.

MAIN FINDING

The normalized 5-1000 Hz motor-on/off variance ratio falls from 8.04 to 1.20. Absolute and relative motor-on PD noise are lower with the rubber-cap-supported mount. The original peak bands are compared without clipping signed excess.

POLARIZATION FINDING

Mean Stokes directions differ by 109.4 degrees on the Poincare sphere. Angular RMS spread is 5.99 degrees before and 6.41 degrees after; mean DoP is 94.30% and 94.46%. Similar DoP does not mean identical polarization direction.

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INTERPRETATION

Consistent with reduced coupling, but mount geometry and optical state both differ. Rubber damping is not isolated from other changes; there is one run per configuration.

MATCHED ACQUISITION

IN2; zero static outputs; 30 s requested per motor state; 5 s settling after motor stop; 15,258.8 samples/s; 16,384 samples per capture; 830 nm PAX setting. Motor off means rotation stopped, not electrical power removed.

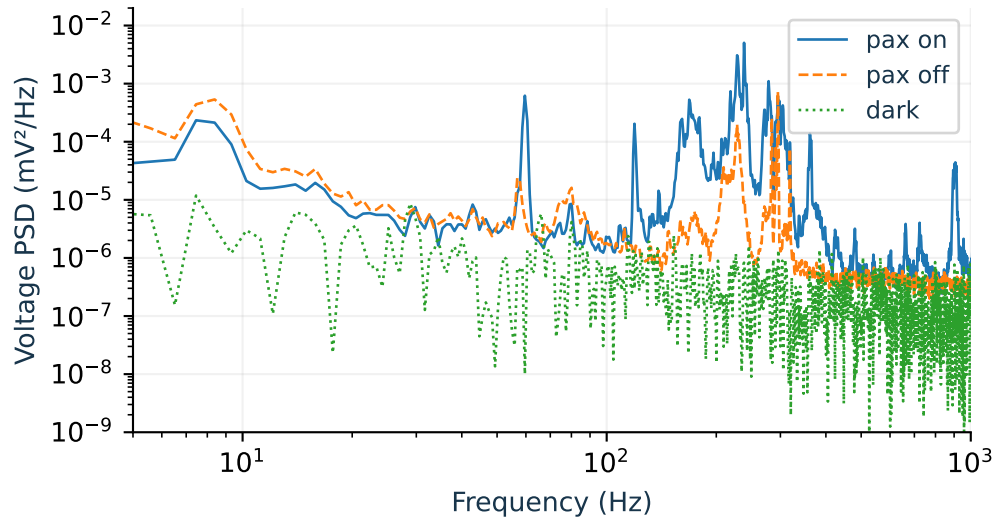
Quantitative comparison

Metric	Before: standard post	After: rubber caps
PD band RMS, motor on	234.2 μ V	34.5 μ V
PD relative band RMS, motor on	2.611%	1.055%
PD band RMS, motor off	79.5 μ V	32.0 μ V
PD relative band RMS, motor off	0.921%	0.962%
Normalized band variance: on / off	8.044	1.202
PD light signal, motor on	8.968 mV	3.266 mV
Mean PAX optical power	589.8 μ W	1064.0 μ W
Mean DoP	94.30%	94.46%
Stokes angular RMS spread	5.99 deg	6.41 deg
Successive-reading angular RMS step	5.01 deg	1.35 deg
PAX power relative RMS	6.97%	1.51%
PD captures, on / off	26 / 28	24 / 27
PAX readings / approximate rate	208 / 6.77 Hz	192 / 6.38 Hz

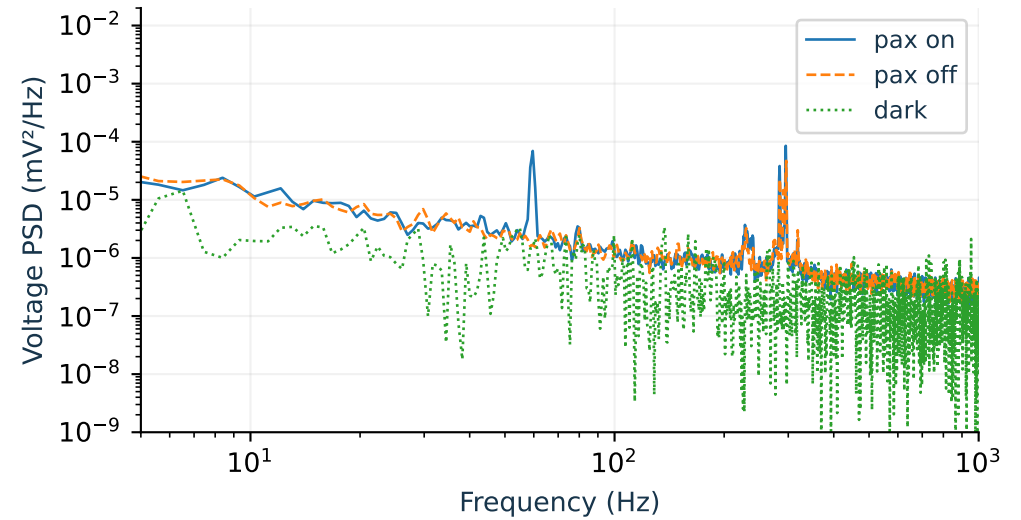
PD metrics use the 5–1000 Hz band. PAX metrics use its slow, motor-on readings only.
 Relative PD power uses the measured dark-corrected mean; voltage noise is not fringe-contrast variance.

PD spectra: suppression of the original peaks

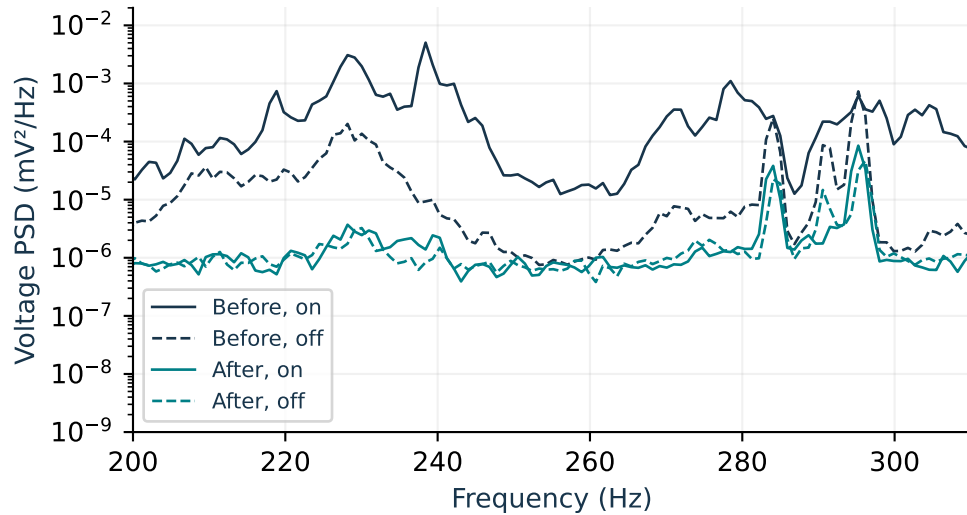
Before: standard post mount



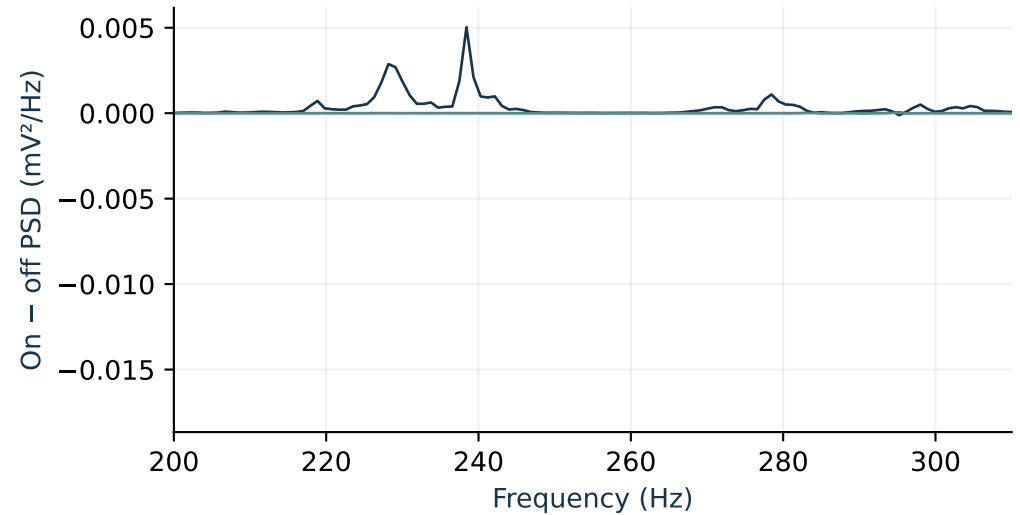
After: rubber-cap support



Same axes: 228 / 238 Hz region

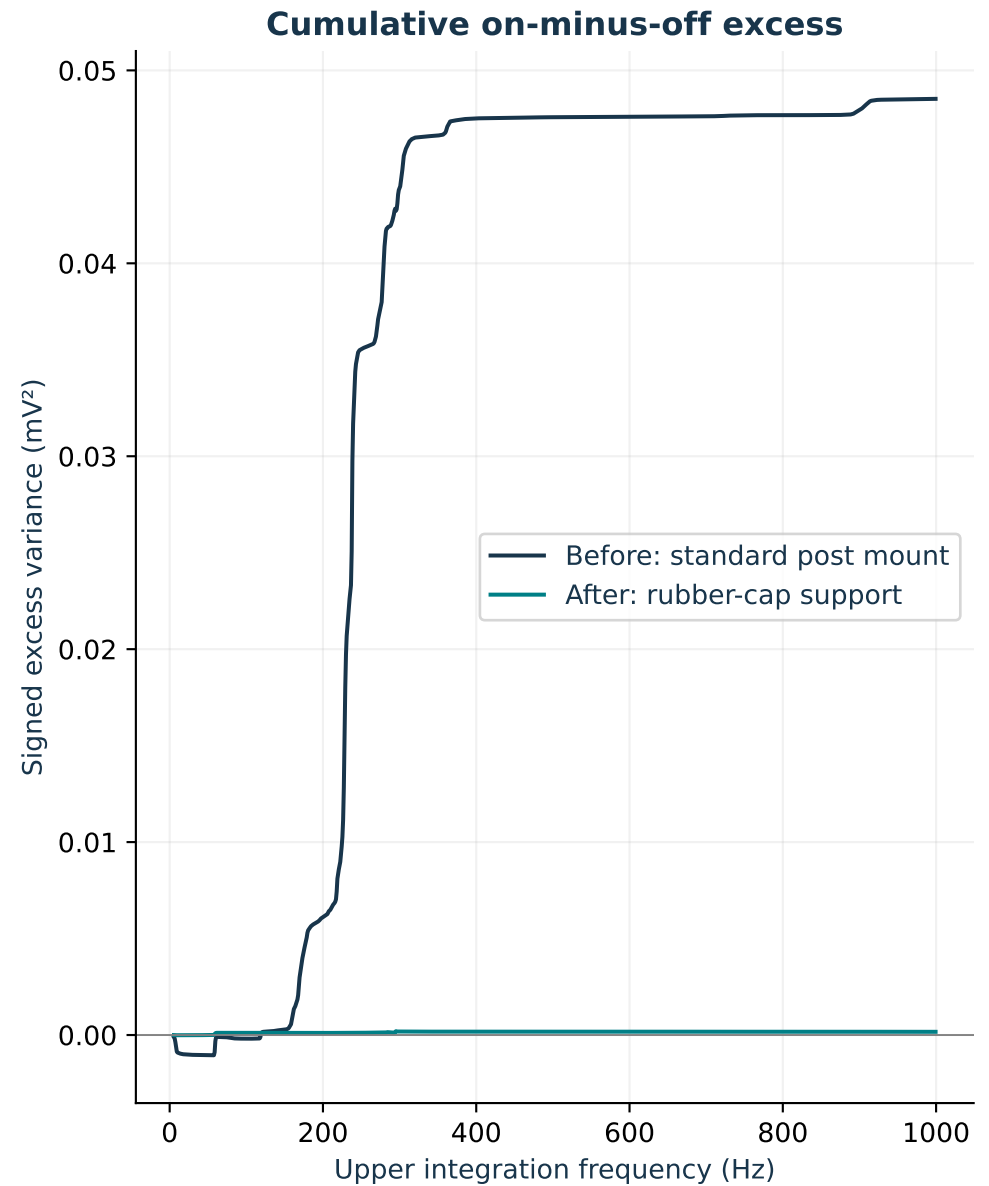
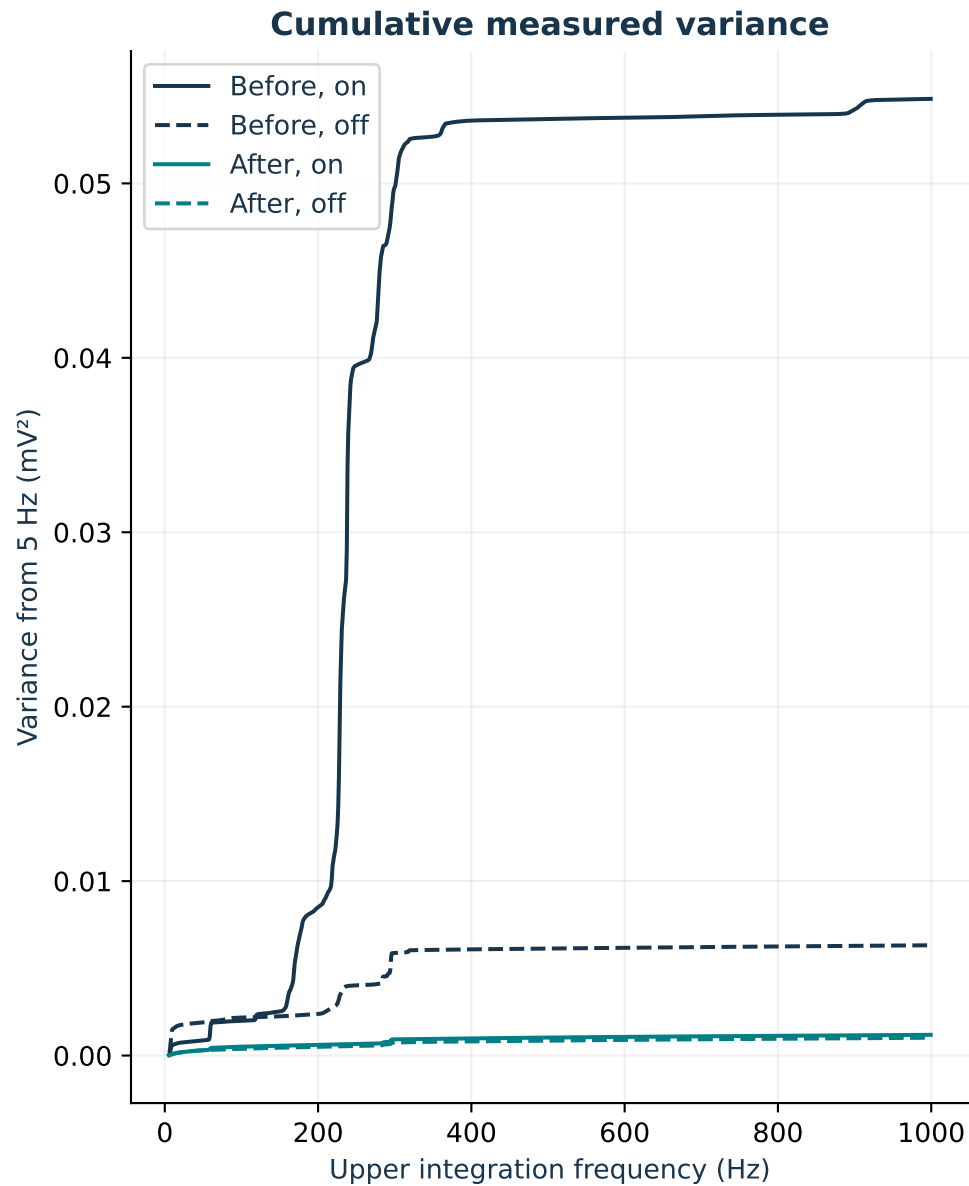


Signed motor-associated excess



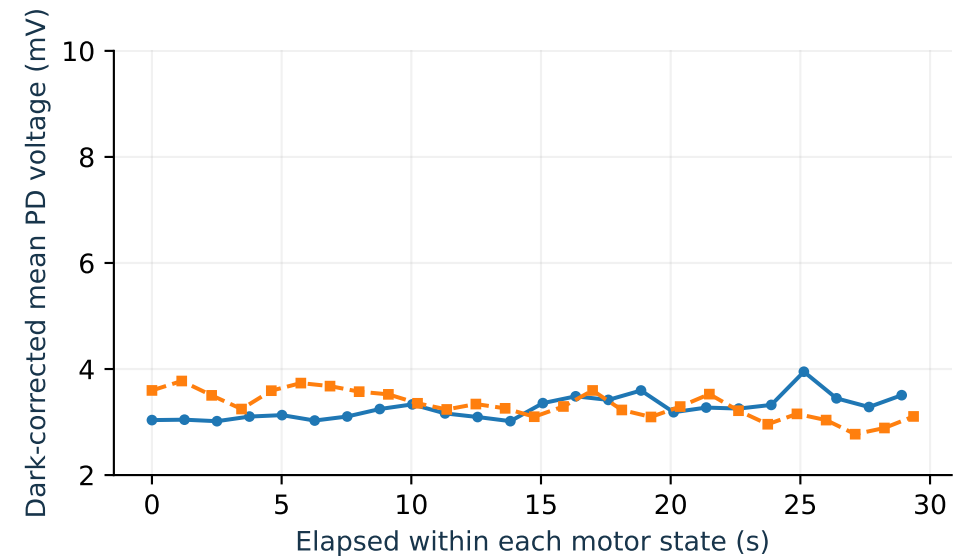
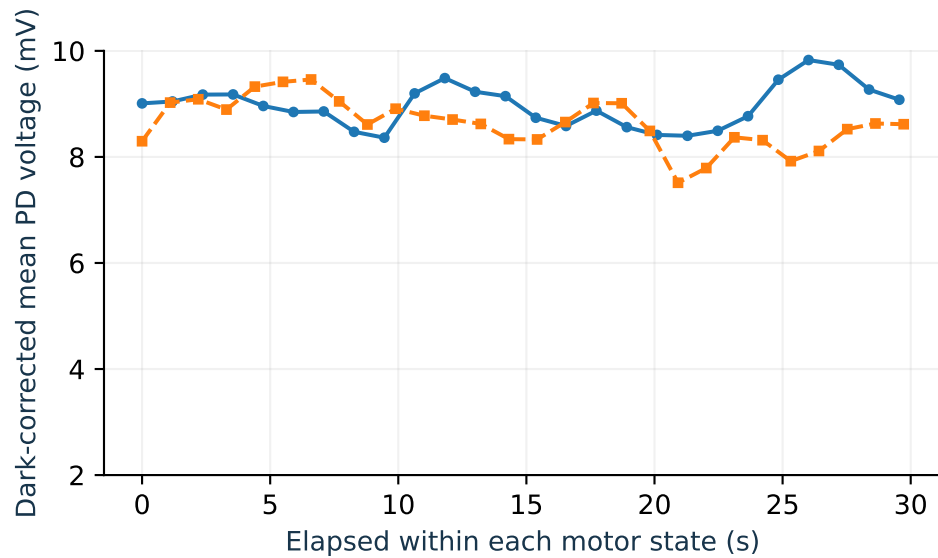
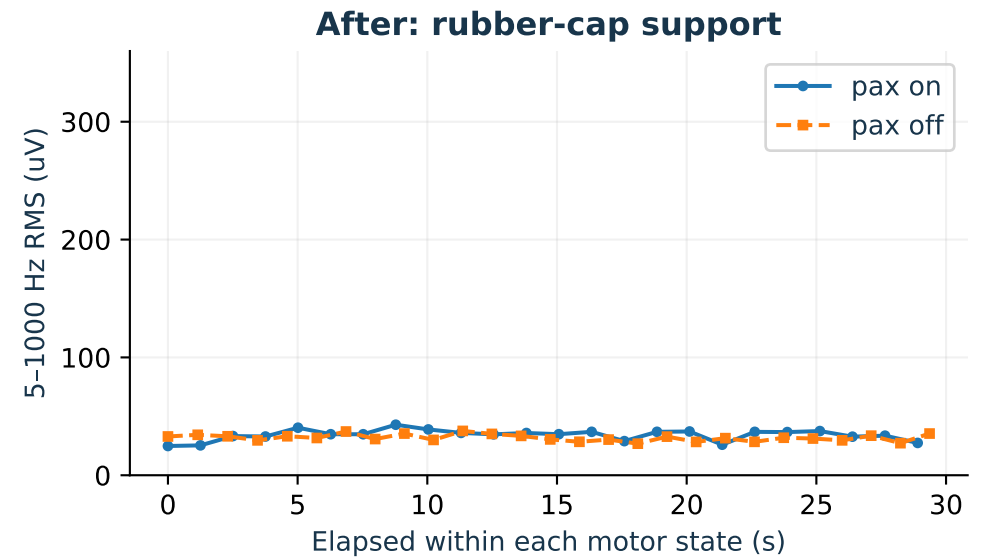
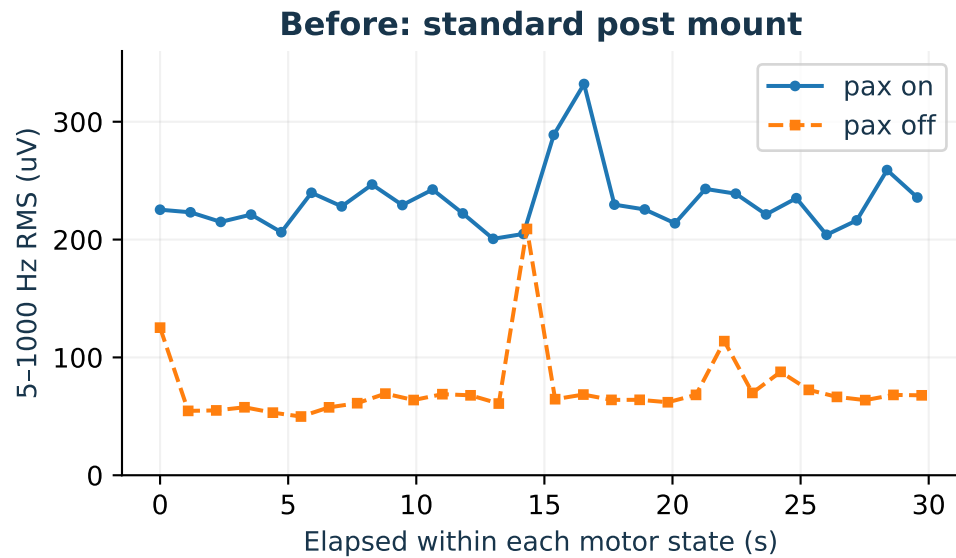
Identical Hann periodogram method and frequency grid (~ 0.93 Hz bins); same vertical scales in the log plots. Dark traces show the electronics baseline. Frequency peaks alone do not identify mechanical causation.

Where the noise power accumulates



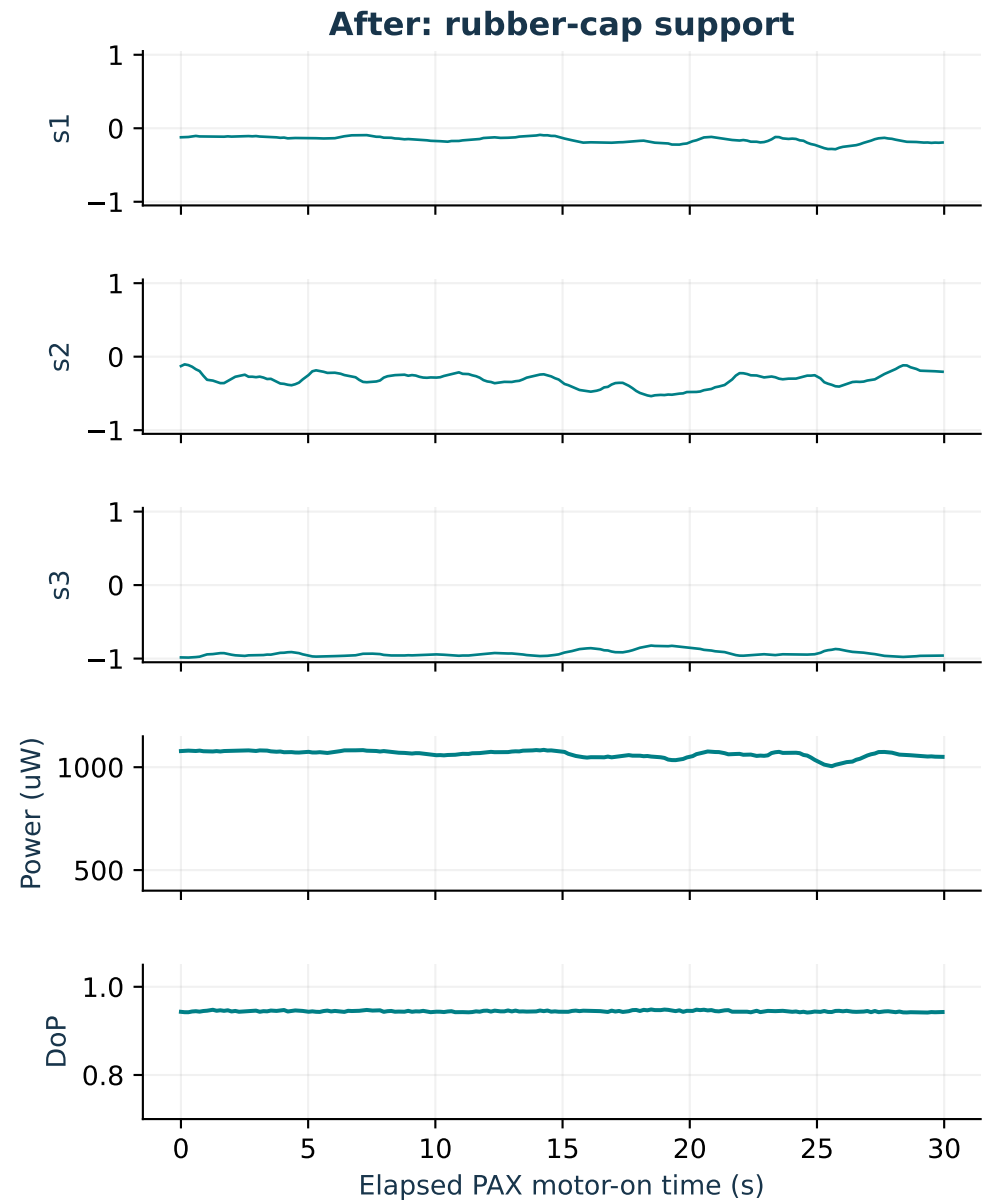
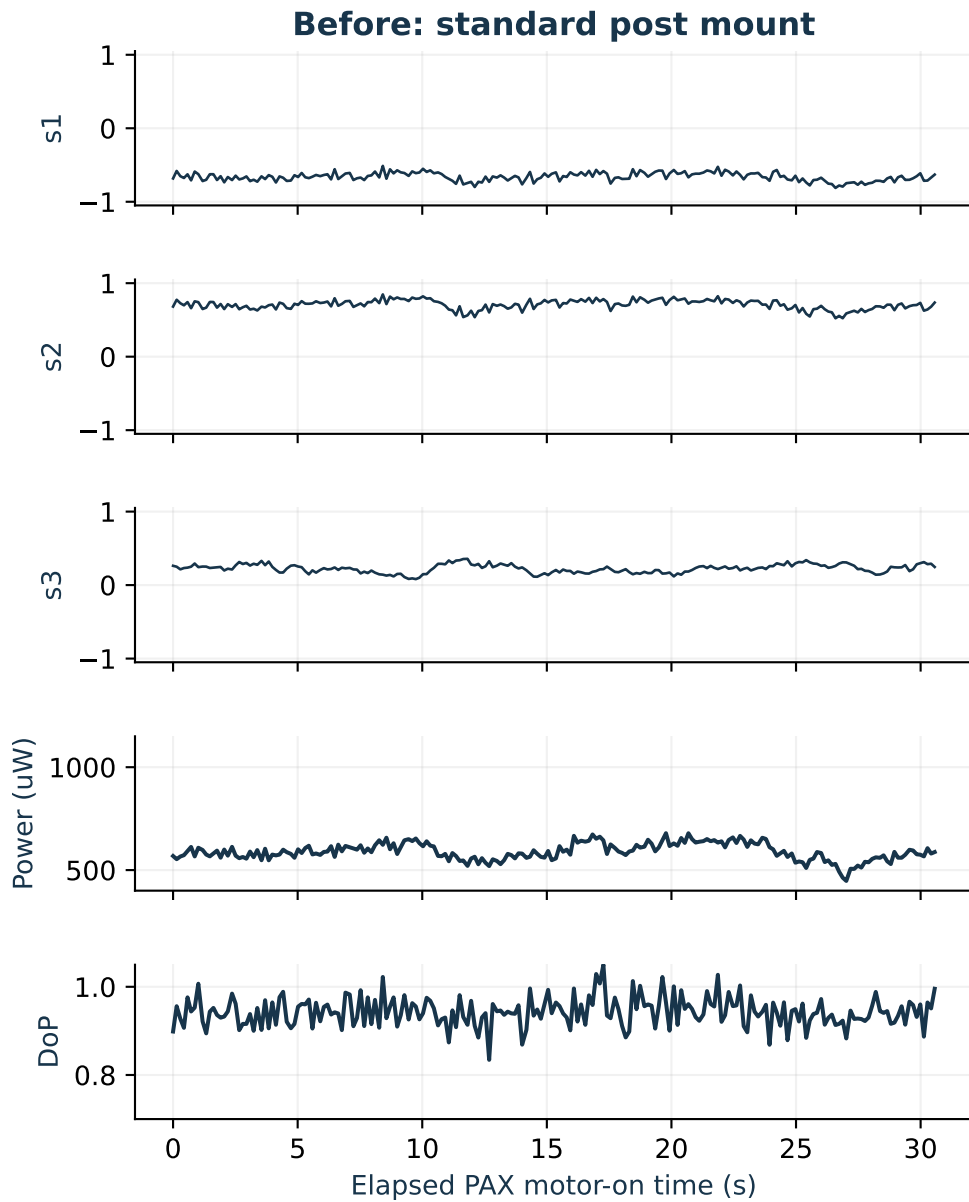
Integration starts at 5 Hz. Sharp rises identify frequency regions contributing most of the noise.
Negative contributions are retained; excess power is not clipped or claimed to be purely mechanical.

PD trace width and light level across captures



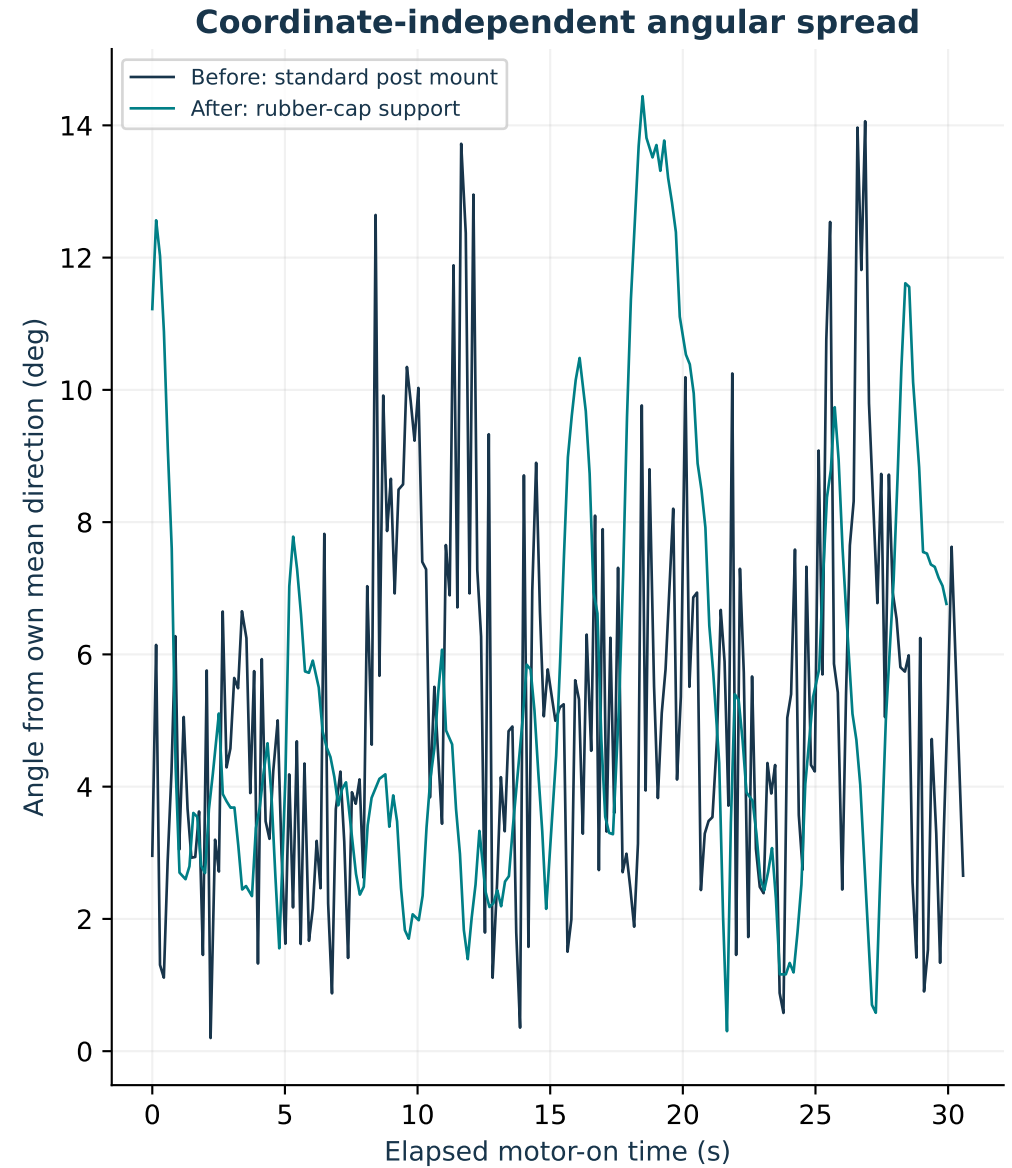
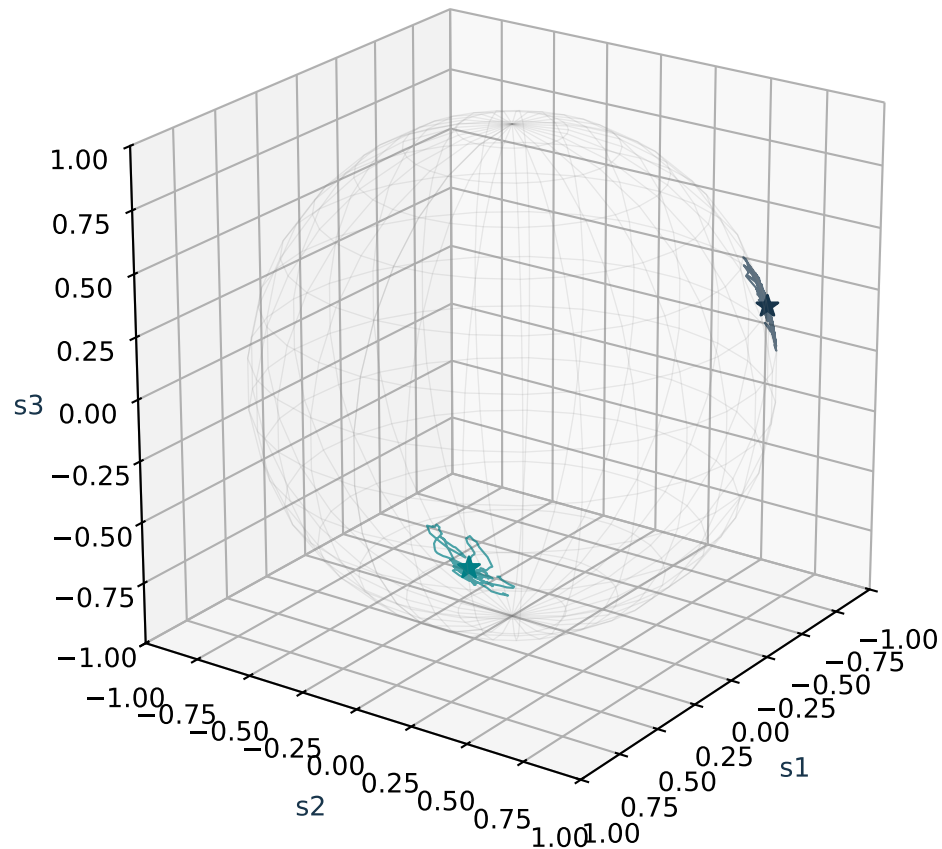
Each point is one ~ 1.074 s FPGA capture, not an independent placement trial. Motor windows are sequential. PD mean light level is lower after the mount change. Compare normalized variance as well as voltage noise.

PAX trajectories: polarization, power and DoP



Normalized Stokes direction is plotted separately from DoP. Identical axis limits enable direct comparison.
PAX samples at $\sim 6.4\text{--}6.8$ Hz: these trajectories cannot resolve the PD peaks near 228 and 238 Hz.

Polarization geometry and angular fluctuations



Mean directions are 109.4° apart; stars mark normalized mean directions, not polarization magnitude.
Angular RMS: 5.99° before, 6.41° after. Slow Stokes spread is distinct from fast PD noise.

Methods, provenance and interpretation

PD CALCULATION

Signed voltages are retained. Each capture is mean-subtracted, Hann-windowed and converted to a one-sided periodogram. Periodograms are averaged within each motor state. Integrate 5–1000 Hz; RMS is the square root of variance. Normalize by the square of each state's dark-corrected mean voltage. No dark-noise PSD subtraction is applied.

PEAK COMPARISON

Integrate signed on-minus-off PSD over ± 3 Hz around the original peak centers. 228.2 Hz: before $1.092\text{e-}08\text{ V}^2$, after $2.972\text{e-}12\text{ V}^2$; 238.4 Hz: before $1.087\text{e-}08\text{ V}^2$, after $5.777\text{e-}12\text{ V}^2$; 278.5 Hz: before $3.785\text{e-}09\text{ V}^2$, after $-6.367\text{e-}13\text{ V}^2$. These are descriptive comparisons, not confidence bounds or mechanical transfer functions; negative excess remains signed.

STOKES CALCULATION

s_1 – s_3 are the suite's normalized Stokes direction; DoP is separate. Mean directions are normalized vector averages. Angular spread is RMS geodesic distance to each run's own mean direction. Successive-step RMS uses adjacent readings at similar, but not identical, update rates. Slow drift and measurement noise are both included.

LIMITS ON ATTRIBUTION

Mount geometry, support and bracing changed together. Optical power, mean polarization and DoP also changed; a different optical operating point can change sensitivity to phase and polarization motion. One run per placement supplies no placement-level replication. Do not describe this as contrast variance or proof of a mechanical frequency.

NEXT DISCRIMINATING MEASUREMENT

Repeat alternating placements or isolation states while preserving optical alignment, polarization operating point, PD input and gain. Record a vibration sensor synchronously if identifying mechanical coupling is the objective.

Methods, provenance and interpretation

VERIFICATION AND REPRODUCIBILITY

Raw CSV values and setup metadata were read directly. Recomputed PD statistics and PAX summary metrics match the saved run summaries to $1e-12$ relative tolerance. Source hashes and software versions are in provenance.json. analyze.py regenerates this report without instrument access; source experiments are not modified.